Linear Response Theory

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Consider a system with a time independent Hamiltonian H_0 and external time-dependant perturbation $H' = -f(t)A$, $f(t)$ is a generalised force (a number) and A is a generalised displacement (an operator). The total Hamiltonian is

$$H(t) = H_0 + H'(t) \tag{1.1}$$

where at time $t = -\infty$, $H = H_0$ and the system is in thermal equilibrium at temperature t. The equilibrium average of some physical quantity B (which could even be A itself, but in general doesn't have to) is

$$\langle B \rangle_0 = \operatorname{tr}\left[\rho_0 B\right] \tag{1.2}$$

where the equilibrium density operator is

$$\rho_0 = \frac{1}{Q_0} e^{-\beta H_0} \tag{1.3}$$

where the partition function is of course

$$Q_0 = \operatorname{tr}\left[e^{-\beta H_0}\right] \tag{1.4}$$

and $\beta = 1/(k_B T)$. Clearly, $\operatorname{tr} \rho_0 = 1$ thus the density operator is normalised. We now consider the time evolution of a general nonequilibrium state $\rho(t)$, which is governed by von Neumann equation

$$i\hbar \frac{\partial \rho(t)}{\partial t} = [H, \rho(t)] \tag{1.5}$$

we now define the Liouvillian as

$$\mathcal{L}(\cdot) \equiv \frac{1}{\hbar} [H, \cdot] \tag{1.6}$$

so the von Neumann equation may be written as

$$\frac{\partial \rho(t)}{\partial t} = -i\mathcal{L}\rho(t) \tag{1.7}$$

where

$$\mathcal{L}(t) = \mathcal{L}_0 + \mathcal{L}'(t) \tag{1.8}$$

each term has a direct correspondence to Eq. (1.1).

Provided that our perturbation is gentle, we expect system's state deviate from the equilibrium state ρ_0 linearly

$$\rho(t) = \rho_0 + \delta\rho(t) \tag{1.9}$$

substitute the two equations above into Eq. (1.7) and provided that the perturbation is sufficiently weak, we may ignore the second order term $\mathcal{L}'\delta\rho(t)$, which gives

$$\frac{\partial\rho}{\partial t} = -i\mathcal{L}_0\rho_0 - i\mathcal{L}_0\delta\rho - i\mathcal{L}'\rho_0 \tag{1.10}$$

where we have suppressed time dependence to reduce clutter on the page. Clearly,

$$\frac{\partial\rho_0}{\partial t} = -i\mathcal{L}_0\rho_0 = 0. \tag{1.11}$$

Therefore

$$\frac{\partial\delta\rho}{\partial t} + i\mathcal{L}_0\delta\rho = -i\mathcal{L}'\rho_0 \tag{1.12}$$

The differential equation is in a form of $y' + P(x)y = Q(x)$, so we can use an integrating factor of the form $\mu(x) = e^{\int P(x)\,dx}$. In our case this integrating factor is $e^{i\mathcal{L}_0 t}$, we multiply the previous equation on both sides by this factor to obtain

$$e^{i\mathcal{L}_0 t}\frac{\partial\delta\rho(t)}{\partial t} + e^{i\mathcal{L}_0 t}i\mathcal{L}_0\delta\rho(t) = -ie^{i\mathcal{L}_0 t}\mathcal{L}'(t)\rho_0 \tag{1.13}$$

$$\frac{\partial}{\partial t}\left(e^{i\mathcal{L}_0 t}\delta\rho(t)\right) = -ie^{i\mathcal{L}_0 t}\mathcal{L}'(t)\rho_0 \tag{1.14}$$

integrate both sides give

$$\int_{t_0}^{t} dt'\,\frac{\partial}{\partial t'}\left(e^{i\mathcal{L}_0 t'}\delta\rho(t')\right) = -i\int_{t_0}^{t} dt'\,e^{i\mathcal{L}_0 t'}\mathcal{L}'(t')\rho_0 \tag{1.15}$$

$$e^{i\mathcal{L}_0 t}\delta\rho(t) - e^{i\mathcal{L}_0 t_0}\delta\rho(t_0) = -i\int_{t_0}^{t} dt'\,e^{i\mathcal{L}_0 t'}\mathcal{L}'(t')\rho_0 \tag{1.16}$$

recall that in our set-up $\lim_{t\to-\infty}\delta\rho(t) = 0$. Therefore,

$$e^{i\mathcal{L}_0 t}\delta\rho(t) = -i\int_{-\infty}^{t} dt'\,e^{i\mathcal{L}_0 t'}\mathcal{L}'(t')\rho_0 \tag{1.17}$$

left multiply both sides by $e^{-i\mathcal{L}_0 t}$

$$\begin{aligned}
\delta\rho(t) &= -i\int_{-\infty}^{t} dt'\,e^{i(t'-t)\mathcal{L}_0}\mathcal{L}'(t')\rho_0 \\
&= -\frac{i}{\hbar}\int_{-\infty}^{t} dt'\,e^{i(t'-t)\mathcal{L}_0}\left[H'(t'),\rho_0\right] \\
&= \frac{i}{\hbar}\int_{-\infty}^{t} dt'\,f(t')e^{i(t'-t)\mathcal{L}_0}\left[A,\rho_0\right] \\
&= \frac{i}{\hbar}\int_{-\infty}^{t} dt'\,f(t')\left[A(t'-t),\rho_0\right]
\end{aligned} \tag{1.18}$$

where we used

$$A(t) = e^{it\mathcal{L}_0}A = e^{itH_0/\hbar}Ae^{-itH_0/\hbar} \tag{1.19}$$

which holds due to H_0 is time independent and ρ_0 does not evolve under it. Now we would like to know the change of a physical observable B. Clearly, the equilibrium average is given by Eq. (1.2). For a general nonequilibrium average

$$\langle B(t)\rangle = \mathrm{tr}\left[\rho(t)B\right] = \langle B\rangle_0 + \mathrm{tr}\left[\delta\rho(t)B\right] \equiv \langle B\rangle_0 + \langle\delta B(t)\rangle \tag{1.20}$$

and using previous result for $\delta\rho(t)$, we have

$$\begin{aligned}\langle \delta B(t)\rangle &= \frac{i}{\hbar}\int_{-\infty}^{t} dt' f(t')\,\mathrm{tr}\,[[A(t'-t),\rho_0]B] \\ &= \frac{i}{\hbar}\int_{-\infty}^{t} dt'\, f(t')\,\mathrm{tr}\,[A(t'-t)\rho_0 B - \rho_0 A(t'-t)B] \\ &= \frac{i}{\hbar}\int_{-\infty}^{t} dt'\, f(t')\Big\langle [B, A(t'-t)]\Big\rangle_0 \end{aligned} \tag{1.21}$$

thus overall

$$\boxed{\langle B(t)\rangle = \langle B\rangle_0 + \frac{i}{\hbar}\int_{-\infty}^{t} dt'\, f(t')\Big\langle [B, A(t'-t)]\Big\rangle_0} \tag{1.22}$$

which is the **Kubo formula**. Let us now define a retarded **linear response function** of variable B with respect to perturbation A with time elapse of τ

$$\phi_{AB}(\tau) \equiv \frac{1}{i\hbar}\Big\langle [A(0), B(\tau)]\Big\rangle_0 = \frac{i}{\hbar}\Big\langle [B(\tau), A]\Big\rangle_0 = \frac{i}{\hbar}\Big\langle [B, A(-\tau)]\Big\rangle_0 \tag{1.23}$$

therefore

$$\langle \delta B(t)\rangle = \int_{-\infty}^{t} dt'\, f(t')\phi_{AB}(t-t') \tag{1.24}$$

notice that the response function ϕ_{AB} is only involves variable A under the time evolution under H_0 and the average is taken in the unperturbed ensemble ρ_0, however, it governs the behaviour of the system away form equilibrium. (See appendix B for an alternative derivation from first order time-dependent perturbation theory)

We shall now return to $\phi_{AB}(t)$ to find out a form that is easier to calculate from theory

$$\begin{aligned}\phi_{AB}(t) &= \frac{1}{i\hbar}\Big\langle [A(0), B(t)]\Big\rangle_0 \\ &= \frac{1}{i\hbar Q_0}\,\mathrm{tr}\left[e^{-\beta H_0}[A(0), B(t)]\right] \\ &= \frac{1}{i\hbar Q_0}\,\mathrm{tr}\left[e^{-\beta H_0}A(0)B(t) - e^{-\beta H_0}B(t)A(0)\right] \\ &= \frac{1}{i\hbar Q_0}\,\mathrm{tr}\left[e^{-\beta H_0}A(0)B(t) - A(0)e^{-\beta H_0}B(t)\right] \\ &= \frac{1}{i\hbar Q_0}\,\mathrm{tr}\left[e^{-\beta H_0}\left(A(0) - e^{\beta H_0}A(0)e^{-\beta H_0}\right)B(t)\right] \\ &= \frac{1}{i\hbar Q_0}\,\mathrm{tr}\left[e^{-\beta H_0}\left(A(0) - A(-i\beta\hbar)\right)B(t)\right] \\ &= \frac{1}{i\hbar Q_0}\,\mathrm{tr}\left[e^{-\beta H_0}\int_{-i\beta\hbar}^{0} d\tau\,\frac{\partial A(\tau)}{\partial\tau}B(t)\right] \\ &= \frac{1}{Q_0}\,\mathrm{tr}\left[e^{-\beta H_0}\int_{0}^{\beta} d\lambda\,\dot{A}(-i\hbar\lambda)B(t)\right] \\ &= \int_{0}^{\beta} d\lambda\,\langle \dot{A}(-i\hbar\lambda)B(t)\rangle_0 \end{aligned} \tag{1.25}$$

We define the equilibrium time correlation function $C_{AB}(t)$ as

$$C_{AB}(t) \equiv \langle A(0)B(t)\rangle_0 \tag{1.26}$$

and the **Kubo-transformed time correlation function** $C^K_{AB}(t)$ as

$$C^K_{AB}(t) \equiv \frac{1}{\beta}\int_0^\beta d\lambda\, \langle A(-i\hbar\lambda)B(t)\rangle_0 \tag{1.27}$$

which can also be written as

$$C^K_{AB}(t) = \frac{1}{\beta}\int_0^\beta d\lambda\, \langle e^{\lambda H_0}A(0)e^{-\lambda H_0}B(t)\rangle_0 = \frac{1}{\beta Q_0}\int_0^\beta d\lambda\ \mathrm{tr}\left[e^{-(\beta-\lambda)H_0}A(0)e^{-\lambda H_0}B(t)\right] \tag{1.28}$$

which seems like the Boltzmann operator $e^{-\beta H_0}$ has been smeared between the two positions in front and behind of $A(0)$. This also leads to a succinct relation between the response function and the Kubo-transformed time correlation function

$$\boxed{\phi_{AB}(t) = \beta C^K_{\dot{A}B}(t)} \tag{1.29}$$

it is easy to show that moving a time derivative from one dynamical variable to another produced a minus sign

$$\boxed{\dot{C}^K_{AB}(t) = C^K_{A\dot{B}}(t) = -C^K_{\dot{A}B}(t)} \tag{1.30}$$

and $C^K_{AB}(t)$ satisfy the following symmetry properties

$$\boxed{C^K_{AB}(t) = C^K_{BA}(-t), \qquad C^K_{AB}(t) = C^K_{BA}(t)^*, \qquad C^K_{AB}(t) = C^K_{AB}(-t)^*}. \tag{1.31}$$

It is the $C^K_{AB}(t)$ we calculate from theory. We now see how this is connected to experiment.

Firstly, notice that since we are considering *linear* response here, as we can generally expand a force in its Fourier series, this suggests that we should consider the system's response to a force oscillating at some well-defined frequency ω. We shall use the following convention for Fourier transforms

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} d\omega\, e^{-i\omega t}\tilde{f}(\omega) \tag{1.32}$$

$$\tilde{f}(\omega) = \int_{-\infty}^{\infty} dt\, e^{i\omega t}f(t) \tag{1.33}$$

Now consider $f(t) = f_0 e^{-i\omega t}$

$$\langle \delta B(t)\rangle = f_0\int_{-\infty}^{t} dt'\, e^{-i\omega t'}\phi_{AB}(t-t') \tag{1.34}$$

now we make the change of variable $\tau = t - t'$

$$\begin{aligned}\langle \delta B(t)\rangle &= -f_0 e^{-i\omega t}\int_{\infty}^{0} d\tau\, e^{i\omega\tau}\phi_{AB}(\tau)\\ &= f_0 e^{-i\omega t}\int_0^{\infty} d\tau\, e^{i\omega\tau}\phi_{AB}(\tau)\end{aligned} \tag{1.35}$$

so system's response looks like the original force $f(t)$, but multiplied by some other function of ω (i.e. the integral over τ part). We now define this as the **generalised susceptibility**

$$\chi_{AB}(\omega) \equiv \int_0^\infty d\tau\, e^{i\omega\tau}\phi_{AB}(\tau) \tag{1.36}$$

crucially, the expression is not quite Fourier transform due to the lower limit only goes down to 0 rather than $-\infty$, this is came from the change of variable earlier, before this change of variable, the upper limit of Eq. (1.24) stops at t rather than ∞. This is due to causality, the system at time t obviously could not respond to the force $f(t')$ at some future time $t' > t$. This has crucial consequences. Furthermore, notice that if we allow ω to be a complex number, the integral only converges in the upper half of the complex plane. (i.e. there is an extra damping factor in the integral that comes from $e^{-\mathrm{Im}(\omega)\tau}$) Subtlety, for real ω, the integral may not converge, but we may take the limit of the imaginary part as it goes to zero ($\mathrm{Im}(\omega) \to 0_+$).

The response of a system with respect to a force of frequency ω can be written most succinctly as the susceptibility times the force

$$\langle \delta B(t) \rangle = \chi_{AB}(\omega) f_0 e^{-i\omega t} \tag{1.37}$$

For a more general force with various frequency components, with contribution from each component as $f_0 = \tilde{f}(\omega)$ we have (also multiply by a harmless factor of 2π to be consistent with our Fourier transform convention)

$$\langle \delta B(t) \rangle = \frac{1}{2\pi}\int_{-\infty}^\infty \chi_{AB}(\omega)\tilde{f}(\omega)e^{-i\omega t} \tag{1.38}$$

using our convention of Fourier transform Eq. (1.27) gives

$$\boxed{\langle \delta \tilde{B}(\omega) \rangle = \chi_{AB}(\omega)\tilde{f}(\omega)} \tag{1.39}$$

Thus, the generalised susceptibility describes how sensitive the system is to a force of some frequency. Thus the central quantity calculated from theory is the Kubo-transformed time correlation function $C^K_{AB}(t)$ and the central quantity measured in experiment is the generalised susceptibility $\chi_{AB}(\omega)$. We will consider a discrete physical case later. But let us continue to develop the general framework for now.

We will now deduce some properties of $\chi_{AB}(\omega)$. Note that $\chi_{AB}(\omega)$ does not have to be real and is in general complex. The real part is an even function of ω:

$$\mathrm{Re}\chi_{AB}(\omega) = \mathrm{Re}\chi_{AB}(-\omega) \tag{1.40}$$

which directly follows from $\cos(\omega t) = \cos(-\omega t)$. The imaginary part is an odd function of ω

$$\mathrm{Im}\chi_{AB}(\omega) = -\mathrm{Im}\chi_{AB}(-\omega) \tag{1.41}$$

which follows from $\sin(\omega t) = -\sin(-\omega t)$. The two symmetries above may be written together as

$$\chi^*_{AB}(\omega) = \chi_{AB}(-\omega) \tag{1.42}$$

We can extend the upper limit of the integral Eq. (1.24) up to ∞ using the Heaviside step function

$$\theta(t) \equiv \begin{cases} 1, & t > 0, \\ 0, & t < 0. \end{cases} \tag{1.43}$$

$$\langle \delta B(t) \rangle = \int_{-\infty}^{\infty} dt'\, f(t') \phi_{AB}(t-t') \theta(t-t') \equiv \int_{-\infty}^{\infty} dt'\, f(t') G_{AB}(t-t') \tag{1.44}$$

so that the integrand still vanishes when $t' > t$. We have also conveniently defined the retarded linear response Green's function $G_{AB}(\tau) \equiv \phi_{AB}(\tau)\theta(\tau)$. So the system's response becomes a convolution of the force with the Green's function. Similarly, we can use the same trick by multiplying the integrand by a step function $\theta(\tau)$ to make it into a Fourier (inverse) transform of the Green's function

$$\chi_{AB}(\omega) = \int_{-\infty}^{\infty} d\tau\, e^{i\omega\tau} \phi_{AB}(\tau)\theta(\tau) = \int_{-\infty}^{\infty} d\tau\, e^{i\omega\tau} G_{AB}(\tau) = \tilde{G}_{AB}(\omega). \tag{1.45}$$

However, we also know from the Convention theorem that

$$\int_{-\infty}^{\infty} dt\, e^{i\omega t} f(t) g(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega'\, \tilde{f}(\omega') \tilde{g}(\omega - \omega') \tag{1.46}$$

therefore

$$\chi_{AB}(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega'\, \tilde{\phi}_{AB}(\omega') \tilde{\theta}(\omega - \omega') \tag{1.47}$$

$\tilde{\phi}_{AB}(\omega)$ is known as **spectral function**. Consider the following representation of the Heaviside theta

$$\theta(t) = \lim_{\epsilon \to 0_+} \frac{1}{2\pi i} \int_{-\infty}^{\infty} d\omega\, \frac{e^{i\omega t}}{\omega - i\epsilon} \tag{1.48}$$

differentiation with respect to t gives

$$\delta(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega\, e^{i\omega t} \tag{1.49}$$

so $\dot{\theta}(t) = \delta(t)$, as expected. (See appendix C for why we need the little term $-i\epsilon$ in the denominator.) Upon change of variable $\omega' = -\omega$, then removing the prime mark afterwards, we have

$$\theta(t) = -\lim_{\epsilon \to 0_+} \frac{1}{2\pi i} \int_{-\infty}^{\infty} d\omega\, \frac{e^{-i\omega t}}{\omega + i\epsilon} \tag{1.50}$$

which of course must satisfy $\dot{\theta}(t) = \delta(t)$ again. From now on taking the limit of $\epsilon \to 0_+$ is left understood, so our formulae looks cleaner. From our convention of Fourier transform we have

$$\boxed{\tilde{\theta}(\omega) = \frac{i}{\omega + i\epsilon}} \tag{1.51}$$

very neat indeed.

Substitute this back into Eq. (1.47) gives

$$\chi_{AB}(\omega) = \frac{i}{2\pi} \int_{-\infty}^{\infty} d\omega' \frac{\tilde{\phi}_{AB}(\omega')}{\omega - \omega' + i\epsilon} = \frac{1}{2\pi i} \int_{-\infty}^{\infty} d\omega' \frac{\tilde{\phi}_{AB}(\omega')}{\omega' - \omega - i\epsilon} \tag{1.52}$$

The $\chi_{AB}(\omega)$ is an analytic thanks to the extra exponentially damping factor of $e^{-\mathrm{Im}(\omega)t}$ definition Eq. (1.36) in the upper half of the complex plane, thus for a closed contour C in this region of analyticity, by Cauchy's theorem, we have

$$\oint_C d\omega\, \chi(\omega) = 0 \tag{1.53}$$

Furthermore, for real ω, and in physical situation, we expect $C^K_{\dot{A}B}(t)$ decays to zero at large τ, the integral also converges. Now consider,

$$\oint_C d\omega' \, \frac{\chi(\omega')}{\omega' - \omega} = 0 \tag{1.54}$$

provided that the contour is in the upper half of the complex plane (may even go along the real axis, but make a tiny clockwise detour in the upper half of the complex plane to avoid the pole at $\omega' = \omega$) We now use a well-known result from complex analysis

$$\chi(\omega) = \frac{1}{\pi i} \mathcal{P} \int_{-\infty}^{\infty} d\omega' \, \frac{\chi(\omega')}{\omega' - \omega} \tag{1.55}$$

where $\mathcal{P}$ denotes Cauchy principal value. It immediately follows that

$$\boxed{\mathrm{Re}\chi(\omega) = \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} d\omega' \, \frac{\mathrm{Im}\chi(\omega')}{\omega' - \omega}, \qquad \mathrm{Im}\chi(\omega) = -\frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} d\omega' \, \frac{\mathrm{Re}\chi(\omega')}{\omega' - \omega}} \tag{1.56}$$

The two equations that relates the real to imaginary parts of the generalised susceptibility, are known as dispersion relations (or **Cramers-Kronig relations**). Furthermore, the pair of equations are also known as Hilbert transforms. An upshot of this is that the susceptibility can neither be purely real nor purely imaginary.

The dispersion relation is a consequence of causality. (i.e. $\chi(\omega)$ is a one-sided Fourier transform of the linear response function $\phi(t)$, the lower limit only goes down to 0 rather than $-\infty$, so $\chi(\omega)$ is analytic in the upper half of the complex plane)

We shall now return to $\phi_{AB}(t)$, we will choose the complete set of eigenstates $\{|n\rangle\}$ (with a set of eigenvalue $\{E_n\}$) of H_0 to evaluate to trace. We also define $\hbar\omega_{nm} \equiv E_n - E_m$ and matrix element $A_{nm} = \langle n|A|m\rangle$. Then it is simple to show that

$$C_{AB}(t) = \sum_{nm} e^{-\beta E_n} A_{nm} B_{mn} e^{-i\omega_{nm} t} \tag{1.57}$$

similarly

$$C_{BA}(t) = \sum_{nm} e^{-\beta E_m} A_{nm} B_{mn} e^{-i\omega_{nm} t} \tag{1.58}$$

therefore

$$\Big\langle [A(0), B(t)] \Big\rangle_0 = C_{AB}(t) - C_{BA}(t) = \sum_{nm} (e^{-\beta E_n} - e^{-\beta E_m}) A_{nm} B_{mn} e^{-i\omega_{nm} t} \tag{1.59}$$

$$\phi_{AB}(t) = \frac{1}{i\hbar} \Big\langle [A(0), B(t)] \Big\rangle_0 = \frac{1}{i\hbar} \sum_{nm} (e^{-\beta E_n} - e^{-\beta E_m}) A_{nm} B_{mn} e^{-i\omega_{nm} t} \tag{1.60}$$

so the spectral function $\tilde{\phi}_{AB}(\omega)$ is

$$\begin{aligned} \tilde{\phi}_{AB}(\omega) &= \frac{1}{i\hbar} \sum_{nm} (e^{-\beta E_n} - e^{-\beta E_m}) A_{nm} B_{mn} \int_{-\infty}^{\infty} dt \, e^{i(\omega - \omega_{nm})t} \\ &= \frac{2\pi}{i\hbar} \sum_{nm} (e^{-\beta E_n} - e^{-\beta E_m}) A_{nm} B_{mn} \delta(\omega - \omega_{nm}) \end{aligned} \tag{1.61}$$

where we recognised the Fourier representation of δ function at the end of the first line. Thus, the spectral function $\tilde{\phi}_{AB}(\omega)$ has peaks at frequencies matching the energy gaps in the spectrum of H_0. It then also immediately follows from Eq. (1.52) that

$$\chi_{AB}(\omega) = \frac{1}{\hbar}\sum_{nm} A_{nm}B_{mn}\frac{e^{-\beta E_n} - e^{-\beta E_m}}{\omega - \omega_{nm} + i\epsilon} \tag{1.62}$$

Furthermore, if we continue from Eq. (1.62)

$$\begin{aligned}\tilde{\phi}_{AB}(\omega) &= \frac{2\pi}{i\hbar}\sum_{nm} e^{-\beta E_n}(1 - e^{\beta\hbar\omega_{nm}})A_{nm}B_{mn}\delta(\omega - \omega_{nm}) \\ &= \frac{2\pi}{i\hbar}\sum_{nm} e^{-\beta E_n}(1 - e^{\beta\hbar\omega})A_{nm}B_{mn}\delta(\omega - \omega_{nm})\end{aligned} \tag{1.63}$$

$$\frac{\tilde{\phi}_{AB}(\omega)}{1 - e^{\beta\hbar\omega}} = \frac{2\pi}{i\hbar}\sum_{nm} e^{-\beta E_n}A_{nm}B_{mn}\delta(\omega - \omega_{nm}) \tag{1.64}$$

then it is simple to show

$$i\hbar\frac{\tilde{\phi}_{AB}(\omega)}{1 - e^{\beta\hbar\omega}} = \int_{-\infty}^{\infty} dt\, e^{i\omega t}C_{AB}(t) = \tilde{C}_{AB}(\omega) \tag{1.65}$$

and by Fourier inversion

$$C_{AB}(t) = \frac{i\hbar}{2\pi}\int_{-\infty}^{\infty}\frac{d\omega\, e^{-i\omega t}\tilde{\phi}_{AB}(\omega)}{1 - e^{\beta\hbar\omega}} \tag{1.66}$$

similarly

$$C_{BA}(t) = \frac{i\hbar}{2\pi}\int_{-\infty}^{\infty}\frac{d\omega\, e^{-i\omega t}e^{\beta\hbar\omega}\tilde{\phi}_{AB}(\omega)}{1 - e^{\beta\hbar\omega}} \tag{1.67}$$

so $C_{AB}(t) - C_{BA}(t) = i\hbar\phi_{AB}(t)$. The two expressions above also allows us to express

$$\begin{aligned}\Big\langle [A(0), B(t)]_+ \Big\rangle_0 &= \frac{i\hbar}{2\pi}\int_{-\infty}^{\infty} d\omega\, e^{-i\omega t}\tilde{\phi}_{AB}(\omega)\left(\frac{1 + e^{\beta\hbar\omega}}{1 - e^{\beta\hbar\omega}}\right) \\ &= \frac{i\hbar}{2\pi}\int_{-\infty}^{\infty} d\omega\, e^{-i\omega t}\tilde{\phi}_{AB}(\omega)\left(\frac{e^{-\beta\hbar\omega/2} + e^{\beta\hbar\omega/2}}{e^{-\beta\hbar\omega/2} - e^{\beta\hbar\omega/2}}\right) \\ &= -\frac{i\hbar}{2\pi}\int_{-\infty}^{\infty} d\omega\, e^{-i\omega t}\tilde{\phi}_{AB}(\omega)\coth\frac{\beta\hbar\omega}{2}\end{aligned} \tag{1.68}$$

where the anti-commutator is defined as $[A, B]_+ \equiv AB + BA$. Notice that the equilibrium average of the anti-commutator between A and B has an extra factor of $-\coth(\beta\hbar\omega/2)$ compared to the commutator version. The anti-commutator equilibrium average can also be equivalently written in terms of Bose-Einstein occupation numbers $\bar{n}(\omega) = (e^{\beta\hbar\omega} - 1)^{-1}$

$$\begin{aligned}\frac{1 + e^{\beta\hbar\omega}}{1 - e^{\beta\hbar\omega}} &= -\frac{2 + e^{\beta\hbar\omega} - 1}{e^{\beta\hbar\omega} - 1} \\ &= -\left(\frac{2}{e^{\beta\hbar\omega} - 1} + 1\right) \\ &= -\left(2\bar{n}(\omega) + 1\right)\end{aligned} \tag{1.69}$$

$$\left\langle [A(0), B(t)]_+ \right\rangle_0 = -\frac{i\hbar}{2\pi} \int_{-\infty}^{\infty} d\omega \, e^{-i\omega t} \tilde{\phi}_{AB}(\omega) \, (2\bar{n}(\omega) + 1) \tag{1.70}$$

Recall that the average energy of a simple harmonic oscillator is $E_\beta(\omega) = (\bar{n}(\omega) + 1/2)\hbar\omega$, which tends to $(1/2)\hbar\omega$ as $\beta \to \infty$ and tends to $k_B T$ as $\beta \to 0$ in accordance with classical equipartition. We now can also write

$$\boxed{\left\langle [A(0), B(t)]_+ \right\rangle_0 = \frac{1}{\pi i} \int_{-\infty}^{\infty} d\omega \, e^{-i\omega t} \tilde{\phi}_{AB}(\omega) \frac{E_\beta(\omega)}{\omega}} \tag{1.71}$$

We now derive a relationship between $\tilde{C}_{AB}(\omega)$ and $\tilde{C}^K_{AB}(\omega)$

$$\begin{aligned}
\tilde{C}^K_{AB}(\omega) &= \int_{-\infty}^{\infty} dt \, e^{i\omega t} C^K_{AB}(t) \\
&= -\int_{-\infty}^{\infty} dt \, e^{i\omega t} \dot{C}^K_{AB}(t) \\
&= -\left[e^{i\omega t} C^K_{AB}(t) \right]_{-\infty}^{\infty} + i\omega \int_{-\infty}^{\infty} dt \, e^{i\omega t} C^K_{AB}(t) \\
&= i\omega \tilde{C}^K_{AB}(\omega)
\end{aligned} \tag{1.72}$$

return to Eq. (1.66)

$$\begin{aligned}
\tilde{C}_{AB}(\omega) &= \frac{i\hbar\beta \tilde{C}^K_{AB}(\omega)}{1 - e^{\beta\hbar\omega}} \\
&= \frac{i\hbar\beta i\omega \tilde{C}^K_{AB}(\omega)}{1 - e^{\beta\hbar\omega}} \\
&= \frac{\beta\hbar\omega}{e^{\beta\hbar\omega} - 1} \tilde{C}^K_{AB}(\omega)
\end{aligned} \tag{1.73}$$

define effective reciprocal temperature as

$$\beta_\omega \equiv \frac{e^{\beta\hbar\omega} - 1}{\hbar\omega} \tag{1.74}$$

which tends to β as $\beta\hbar\omega \to 0$, thus we obtain the relationship (in our Fourier transform convention)

$$\boxed{\tilde{C}^K_{AB}(\omega) = \frac{\beta_\omega}{\beta} \tilde{C}_{AB}(\omega)} \tag{1.75}$$

We now return to $\chi_{AB}(\omega)$. We already know that $\chi_{AB}(\omega)$ is an an analytic function in the upper half of the complex ω-plane, thus the susceptibility of physical (real) ω are limits down unto the real axis. In other words, we can define an analytic function (excluding the real axis)

$$K_{AB}(\omega) \equiv \frac{1}{2\pi i} \int_{-\infty}^{\infty} d\omega' \, \frac{\tilde{\phi}_{AB}(\omega')}{\omega' - \omega} \tag{1.76}$$

and the *retarded* (physical) susceptibility is

$$\chi_{AB}(\omega) = \chi^+_{AB}(\omega) = K_{AB}(\omega + i\epsilon) \tag{1.77}$$

and the *advanced* susceptibility is

$$\chi^{-}_{AB}(\omega) = K_{AB}(\omega - i\epsilon) \tag{1.78}$$

which does not have to be the same as the retarded version. In fact, we can work out the difference between the two

$$\chi^{+}_{AB}(\omega) - \chi^{-}_{AB}(\omega) = \frac{1}{2\pi i}\int_{-\infty}^{\infty} d\omega'\, \tilde{\phi}_{AB}(\omega')\left(\frac{1}{\omega' - (\omega + i\epsilon)} - \frac{1}{\omega' - (\omega - i\epsilon)}\right) \tag{1.79}$$

We shall now use the Sokhotski-Plemelj identity

$$\lim_{\epsilon \to 0_+} \frac{1}{x \pm i\epsilon} = \mathcal{P}\frac{1}{x} \mp i\pi\delta(x) \tag{1.80}$$

therefore

$$\lim_{\epsilon \to 0_+}\left(\frac{1}{x - i\epsilon} - \frac{1}{x + i\epsilon}\right) = 2\pi i\delta(x) \tag{1.81}$$

and

$$\boxed{\chi^{+}_{AB}(\omega) - \chi^{-}_{AB}(\omega) = \tilde{\phi}_{AB}(\omega)} \tag{1.82}$$

So the physical retarded generalised susceptibility $\chi_{AB}(\omega)$ is the limit of the master analytic function $K_{AB}(\omega)$ from the upper half of the complex plane down onto the real axis, and the spectral function $\tilde{\phi}_{AB}(\omega)$ is the difference of the discontinuity from above the below the real axis of this master analytic function. Relating $\tilde{\phi}_{AB}(\omega)$ back to $\tilde{C}^{K}_{AB}(\omega)$ gives

$$\chi_{AB}(\omega) = \frac{\beta}{2\pi}\int_{-\infty}^{\infty} d\omega'\, \frac{\omega'}{\omega' - \omega - i\epsilon}\tilde{C}^{K}_{AB}(\omega') \tag{1.83}$$

using Sokhotski-Plemelj identity again yields

$$\boxed{\mathrm{Re}\chi_{AB}(\omega) = \frac{\beta}{2\pi}\mathcal{P}\int_{-\infty}^{\infty} d\omega'\, \frac{\omega'}{\omega' - \omega}\tilde{C}^{K}_{AB}(\omega')} \tag{1.84}$$

$$\boxed{\mathrm{Im}\chi_{AB}(\omega) = \frac{\beta\omega}{2}\tilde{C}^{K}_{AB}(\omega)} \tag{1.85}$$

These two equation relate the key experimental quantity $\chi_{AB}(\omega)$ is related to the key theoretical quantity $\tilde{C}^{K}_{AB}(\omega)$. These equations are also fluctuation-dissipation relation where the fluctuation at equilibrium is contained in $C^{K}_{AB}(t)$ and the dissipation when the system is away from equilibrium is contained in $\chi_{AB}(\omega)$.

Appendix A

In classical statistical mechanics the state of the system is expressed as a probability distribution $\rho(\mathbf{p}, \mathbf{q})$, where $\mathbf{q} = (q_1, \ldots, q_N)$ are positions of N degrees of freedom and $\mathbf{p}$ is the conjugate momenta. The probability in some small phase space (hyper) volume $d\tau$ is $\rho(\mathbf{p}, \mathbf{q})d\tau$. The time evolution of the system is governed by Hamilton's equations

$$\dot{\mathbf{q}} = \frac{\partial H}{\partial \mathbf{p}}, \qquad \dot{\mathbf{p}} = -\frac{\partial H}{\partial \mathbf{q}} \tag{A1}$$

we now define the Poisson bracket between two dynamical variables F and G as

$$\{F, G\} \equiv \sum_i \left(\frac{\partial F}{\partial q_i} \frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial p_i} \frac{\partial G}{\partial q_i} \right) = \frac{\partial F}{\partial \mathbf{q}} \cdot \frac{\partial G}{\partial \mathbf{p}} - \frac{\partial F}{\partial \mathbf{p}} \cdot \frac{\partial G}{\partial \mathbf{q}}. \tag{A2}$$

By the conservation of probability we have

$$\begin{aligned} 0 &= \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial \mathbf{q}} \cdot (\dot{\mathbf{q}}\, \rho) + \frac{\partial}{\partial \mathbf{p}} \cdot (\dot{\mathbf{p}}\, \rho) \\ &= \frac{\partial \rho}{\partial t} + \frac{\partial H}{\partial \mathbf{p}} \cdot \frac{\partial \rho}{\partial \mathbf{q}} - \frac{\partial H}{\partial \mathbf{q}} \cdot \frac{\partial \rho}{\partial \mathbf{p}} \\ &= \frac{\partial \rho}{\partial t} + \{\rho, H\}. \end{aligned} \tag{A3}$$

The first line is in fact continuity but written in a form for phase space, we used the Hamilton's equation on the second line and the definition of the Poisson bracket on the third line. Define the classical Liouvillian as

$$\mathcal{L}(\cdot) \equiv i\{H, \cdot\} \tag{A4}$$

Eq. (A3) may be written as

$$\frac{\partial \rho(t)}{\partial t} = -i\mathcal{L}\rho(t) \tag{A5}$$

which is Liouville equation, with formal solution

$$\rho(t) = e^{-i\mathcal{L}t}\rho(0) \tag{A6}$$

Appendix B

We show an alternative derivation of Kubo formula here. The time evolution of a pure state is given by the time-dependent Schrödinger equation (TDSE)

$$i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} = H(t)|\psi(t)\rangle \tag{B1}$$

define an unitary time-evolution $U(t, t_0)$ via

$$U(t, t_0)|\psi(t_0)\rangle = |\psi(t)\rangle \tag{B2}$$

so that TDSE can be rewritten as

$$i\hbar \frac{\partial U(t, t_0)}{\partial t} = H(t)U(t, t_0) \tag{B3}$$

we can recover Eq. (B1) by multiplying both sides by $|\psi(t_0)\rangle$. If $H(t) = H_0$ (i.e. is time independent) then we have

$$U_0(t) = e^{-itH_0/\hbar} \tag{B4}$$

however, we are interested in a case where $H(t)$ have the form

$$H(t) = H_0 - F(t)A \tag{B5}$$

where $F(t)$ is a generalised force (a real number) and A is an observable (a Hermitian operator). Consider the following Ansatz

$$U(t,t_0) = U_0(t)U'(t,t_0) \tag{B6}$$

substitute back into Eq. (B3) yields

$$i\hbar\frac{\partial U_0(t)}{\partial t}U'(t,t_0) + i\hbar U_0(t)\frac{\partial U'(t,t_0)}{\partial t} = (H_0 - F(t)A)U_0(t)U'(t,t_0) \tag{B7}$$

notice that the first term on both sides cancels, so we have

$$i\hbar U_0(t)\frac{\partial U'(t,t_0)}{\partial t} = -F(t)AU_0(t)U'(t,t_0) \tag{B8}$$

$$i\hbar\frac{\partial U'(t,t_0)}{\partial t} = -F(t)U_0^\dagger(t)AU_0(t)U'(t,t_0) \tag{B9}$$

where we have used the unitarity of $U_0(t)$ and $F(t)$ is a number, which commutes with all operators, we now define the Heisenberg time evolved operator under H_0

$$A(t) = U_0^\dagger(t)AU_0(t) \tag{B10}$$

therefore

$$i\hbar\frac{\partial U'(t,t_0)}{\partial t} = -F(t)A(t)U'(t,t_0) \tag{B11}$$

$$\frac{\partial U'(t,t_0)}{\partial t} = -\frac{1}{i\hbar}F(t)A(t)U'(t,t_0) \tag{B12}$$

then integrating both sides of Eq. (B12) gives

$$U'(t,t_0) - U'(t_0,t_0) = -\frac{1}{i\hbar}\int_{t_0}^{t} dt'\, F(t')A(t')U'(t',t_0) \tag{B13}$$

we now provide the boundary condition

$$U'(t_0,t_0) = 1 \tag{B14}$$

and slight rearrangement gives

$$U'(t,t_0) = 1 - \frac{1}{i\hbar}\int_{t_0}^{t} dt'\, F(t')A(t')U'(t',t_0) \tag{B15}$$

so $U'(t,t_0)$ depends on itself in an iterative sense, so it is still complicated to solve, however, here we focus on a simpler situation, where the external perturbation is gentle, and the system's response is linear (i.e. we only focus on the first order solution) This means we substitute $U'(t',t_0) = 1$ in the integral on the right-hand side, therefore to first order we have

$$U'(t,t_0) \simeq 1 - \frac{1}{i\hbar}\int_{t_0}^{t} dt'\, F(t')A(t') \tag{B16}$$

and full time-evolution operator (to first order) from our original Ansatz Eq. (B6)

$$U(t,t_0) = U_0(t)\left[1 - \frac{1}{i\hbar}\int_{t_0}^{t} dt'\, F(t')A(t')\right] \tag{B17}$$

we consider the case of where the perturbation is switched on at $t_0 = -\infty$, and simply the notation slightly $U(t, -\infty) = U(t)$

$$U(t) = U_0(t)\left[1 - \frac{1}{i\hbar}\int_{-\infty}^{t} dt'\, F(t')A(t')\right] \tag{B18}$$

Therefore the time-evolution of some dynamical variable B (which could even be A itself) is

$$\begin{aligned} B_{\text{tot}}(t) &= U^\dagger(t)BU(t) \\ &= \left[1 + \frac{1}{i\hbar}\int_{-\infty}^{t} dt'\, F(t')A(t')\right] U_0^\dagger(t)BU_0(t)\left[1 - \frac{1}{i\hbar}\int_{-\infty}^{t} dt'\, F(t')A(t')\right] \\ &= \left[1 + \frac{1}{i\hbar}\int_{-\infty}^{t} dt'\, F(t')A(t')\right] B(t)\left[1 - \frac{1}{i\hbar}\int_{-\infty}^{t} dt'\, F(t')A(t')\right] \\ &= B(t) + \frac{1}{i\hbar}\int_{-\infty}^{t} dt'\, F(t')[A(t'), B(t)] \end{aligned} \tag{B19}$$

finally we average $\langle \cdots \rangle_0$ under the equilibrium ensemble

$$\langle \cdots \rangle_0 \equiv \frac{1}{Q_0}\,\text{tr}\left[e^{-\beta H_0}\cdots\right] \tag{B20}$$

where the partition function is $Q_0 = \text{tr}\left[e^{-\beta H_0}\right]$, therefore

$$\langle B_{\text{tot}}(t)\rangle_0 = \langle B(t)\rangle_0 + \frac{1}{i\hbar}\int_{-\infty}^{t} dt'\, F(t')\Big\langle [A(t'), B(t)]\Big\rangle_0 \tag{B21}$$

notice that since $[U_0(t), \rho_0] = 0$, we have $\langle B(t)\rangle_0 = \langle B\rangle_0$, we then relabel $\langle B_{\text{tot}}(t)\rangle_0 = \langle B(t)\rangle$ and define the linear response function as

$$\phi_{AB}(t) \equiv \frac{1}{i\hbar}\Big\langle [A(0), B(t)]\Big\rangle_0 \tag{B22}$$

it is simple to show that

$$\phi_{AB}(t - t') = \frac{1}{i\hbar}\Big\langle [A(t'), B(t)]\Big\rangle_0 \tag{B23}$$

at last

$$\langle B(t)\rangle = \langle B\rangle_0 + \int_{-\infty}^{t} dt'\, F(t')\phi_{AB}(t - t') \tag{B24}$$

which is Kubo formula.

Appendix C

Heaviside step function is defined as

$$\theta(t) \equiv \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases} \tag{A1}$$

lets find an integral representation of this. Consider,

$$I_\epsilon(t) \equiv \frac{1}{2\pi i}\int_{-\infty}^{\infty} d\omega\, \frac{e^{i\omega t}}{\omega - i\epsilon}, \qquad \epsilon > 0 \tag{A2}$$

First consider $t > 0$, we close the contour using a very large semicircle on the upper half of the complex plan the extra damping factor $e^{-\mathrm{Im}(\omega)t}$ in the integral converges the integral. The semicirlce contains a pole at $\omega = i\epsilon$. By Jordan's lemma the integral along the curved part above the real axis on the semicircle vanishes. Then by Cauchy residue theorem, we have,

$$I_\epsilon(t) = e^{i(i\epsilon)t} = e^{-\epsilon t} \tag{A3}$$

taking the limit of both sides from above gives,

$$\lim_{\epsilon\to 0_+} I_\epsilon(t) = 1, \qquad t > 0. \tag{A4}$$

For negative t, we close the contour on the bottom half of the complex plane to converge the integral and again use Jordan's lemma, however, the contour now does not contain a pole, by Cauchy theorem,

$$\lim_{\epsilon\to 0_+} I_\epsilon(t) = 0, \qquad t < 0. \tag{A5}$$

thus we found an integral representation of the step function,

$$\boxed{\theta(t) = \lim_{\epsilon\to 0_+} \frac{1}{2\pi i}\int_{-\infty}^{\infty} d\omega\, \frac{e^{i\omega t}}{\omega - i\epsilon}} \tag{A6}$$